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Object-aware semantic mapping using probability density functions for indoor relocalization and path planning

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As indoor robots are expected to operate in increasingly complex environments, the need for rich and scalable semantic representations has become critical. While semantic mapping is a standard tool, existing representations often fall at two extremes: dense voxel-based maps that are computationally expensive to query, or schematic graphs that lack geometric detail. This trade-off limits the scalability of semantic maps in real-world tasks. We propose an object-aware semantic mapping framework that models key static objects using probability density functions (PDFs). Objects like beds, fridges or desks are detected via 3D point cloud processing and encoded as 2D probabilistic occupancy distributions. This formulation provides a compact, robust representation that preserves semantic identity and geometric shape, while handling noise and partial views. The map is structured around rooms and their contents, enabling global relocalization and semantically informed path planning. Using Differential Evolution and Kullback-Leibler divergence, our method achieves robust relocalization without prior pose. Its object-centric, probabilistic nature also supports functional scene understanding for context-aware navigation. We validate our approach on a benchmark dataset and in a real apartment, showing improved performance over traditional methods in ambiguous or cluttered scenes, and demonstrating the advantages of a unified representation for multiple robotic behaviors.

Keywords Object awareness, Semantic mapping, Probability density functions, Semantic relocalization, Path planning

The ability to build and exploit semantic maps is essential for mobile robots operating in indoor environments. While geometric representations provide the spatial structure required for navigation, they lack the semantic richness necessary for higher-level reasoning. Semantic maps attempt to bridge this gap by associating symbolic meaning to spatial data. However, most existing approaches fall at two extremes: dense voxel-based maps or annotated schematic models. In practice, voxel maps often scale poorly, leading to large storage requirements; their discretized structure can also result in unnatural, grid-like paths in complex environments, and their integration with higher-level semantic layers remains challenging[1]. Schematic models like topological maps are easier to query but discard geometric fidelity[2]. These trade-offs limit their applicability for real-world tasks, especially those requiring both geometric precision and semantic understanding.

In this work, we propose a unified, object-centric semantic mapping framework that models key static objects as probability density functions (PDFs) over the 2D floor plane. This compact representation captures both the shape and semantic identity of objects from 3D observations, enabling robust, geometry-informed semantic reasoning. By organizing semantic information into a layered 2D structure, rather than storing full volumetric representations, our map achieves balance: it preserves sufficient geometric fidelity for reasoning while remaining easily accessible. In this sense, we address the efficiency of the map representation itself, independent of the runtime of downstream tasks such as global localization. Moreover, since objects are modeled as probability functions, the information can be exploited in a continuous manner, which naturally avoids the rigid, stepped paths often associated with discrete voxel grids or topological representations.

The map is organized in a room-centric fashion, where each room is described by its enclosing walls and the PDF-modeled labeled objects it contains, with each object label maintained in a separate layer to avoid overwriting information. This representation reflects the way humans intuitively understand indoor environments: rooms

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are typically defined not only by their geometry but by the objects they contain. For example, a refrigerator or stove implies a kitchen, while a bed suggests a bedroom. By anchoring spatial understanding around function-defining objects, our map supports a functional interpretation of the environment. We demonstrate the map's direct applicability in two distinct robotic tasks without requiring task-specific adaptations: semantic global relocalization and object-aware path planning. This dual use showcases the versatility and reusability of the map for robot navigation applications.

Global relocalization allows robots to identify their location on a known map without prior pose information. This ability is necessary when a robot first starts working in an environment and needs to establish its initial location. Despite significant advances in robotic navigation, conventional relocalization techniques based on geometric data, such as LiDAR measurements[3,4], exhibit inherent limitations like symmetrical geometries or reflective surfaces that can result in ambiguous pose estimates. Building on the idea that the function of a room is determined by the objects it contains[5], the proposed relocalization method focuses on identifying the most characteristic objects on the environment. This idea has been proven to be efficient in other applications like object searching[6,7]. We define an optimization algorithm based on Differential Evolution (DE) for robot relocalization.

Semantically informed path planning allows robots to reason about their surroundings and adapt their trajectories according to the meaning and function of different areas (e.g., avoiding bedrooms at night or targeting tables for object search). Traditional approaches often rely on manually defined semantic zones or hard-coded costmaps, which lack adaptability and scalability[8]. In contrast, our map enables continuous, object-aware path planning using PDFs as smooth spatial influences.

The main contributions of this work are:

- We introduce a compact semantic map based on probabilistic encodings of static, function-defining objects, enabling persistent and human-like room understanding.
- We promote map reusability across robotic tasks like relocalization and semantic path planning by modeling object observations as continuous probability functions that unify semantic content and spatial geometry, eliminating the need for transformations between symbolic and metric domains.
- We introduce a novel relocalization method that integrates semantic perception with evolutionary optimization, aligning object-centric observations to a global map using Differential Evolution and Kullback-Leibler (KL) divergence for efficient and accurate global pose estimation.
- We enable context-aware trajectory generation by guiding navigation through semantically meaningful regions, prioritizing or avoiding areas based on task context, using smooth object likelihood distributions that allow gradual influence over path shapes.

To validate the proposed framework, we perform a comprehensive evaluation in two different scenarios: a public indoor benchmark dataset and a real-world furnished apartment. Results demonstrate that our probabilistic, object-centric representation not only supports high-level biased path planning behaviors, but also improves localization robustness in perceptually ambiguous scenes. Crucially, the unified map structure enables these capabilities without requiring multiple representations, task-specific tuning or complex symbolic reasoning pipelines.

Related work

Semantic object-based maps in indoor environments

A variety of object-based maps can be found in literature. Representing objects with a 2D point, as proposed in[9,10], offers simplicity and computational efficiency but at the cost of losing essential information like shape, size, and orientation, which is important for localization as it can help to disambiguate between areas. Using 3D bounding boxes[11,12] better preserves position and size, but introduces orientation ambiguity and oversimplifies geometric features. This is particularly problematic for complex or irregularly shaped objects. Moreover, these oversimplified representations struggle to provide sufficient detail for fine-grained spatial analysis.

Similar to[13,14], we propose projecting 3D object points onto the 2D map plane to reduce data dimensionality while retaining information on position, orientation, and shape. This approach achieves a balance between computational efficiency and representational richness, enabling the preservation of geometric features without the overhead associated with full 3D data processing. Furthermore, we propose an improvement over the state of the art by incorporating a probabilistic treatment using PDFs to account for detection uncertainty. This representation also aligns with recent trends in indoor layout reconstruction, such as[15], which utilizes denoised density maps and diffusion models for structural modeling from panoramic inputs.

Semantic localization in indoor environments

Semantic localization primarily relies on camera observations to identify specific elements or features within images that enhance robot positioning. Recent works such as[16–19] use features for visual localization due to their robust performance under transformations, given their scale and rotation invariance. In[20], line segments are additionally detected, providing a complementary geometric representation that captures structural details. Authors in[21] employ convolutional neural networks to extract unit feature vectors from images to characterize locations. Despite their effectiveness and speed, these methods are not intuitive for users coexisting with robots, limiting their applicability.

Similarly, recent research explores the use of implicit representations. The work in[22] proposes a neural fields-based mapping approach, where a mapping network learns the signed distance field of the scene. Authors in[23] propose multi-scale axis-aligned feature planes to extract scene features. In[24], the proposed method incorporates a hierarchical implicit semantic representation to fuse several scene features used for localization. A neural

implicit approach is proposed in[25], where multi-level local information is included. Although these methods significantly advance the ability to represent complex scenarios for localization, they entail high computational costs associated with constructing and querying these representations. Another main drawback is the complexity of integrating these representations with other map types, as their representation is implicit.

Other works propose landmarks similar to those used by people. Authors in[26] extract heatmaps on images corresponding to visually significant regions. However, there is a lack of correspondence between the 2D image and the 3D space. In[27], three landmark types are used: single-object landmarks, multiple-object landmarks and scene landmarks, like corners or intersections. However, this abstraction sacrifices detailed spatial characteristics, limiting their precision in complex environments.

Path planning with semantic annotations

Recent research has explored the integration of semantic information into path planning to enable more context-aware navigation. Traditional approaches such as[28] enrich grid-based maps with object graphs. Other works like[8] operate over semantic maps derived from projecting 3D information directly into a grid map. However, these maps are discrete and they do not capture uncertainty in object detection or location.

To incorporate semantic information more adaptively, some authors have proposed probabilistic approaches[29]. develop a perception-based planner over uncertain semantic maps. However, their method does not address the direct generation of probabilistic distributions from sensor observations. Moreover, geometric object uncertainty is modeled using a Gaussian distribution, assuming that each object is spatially compact, symmetric and ellipsoidal in shape, which limits its ability to represent irregular, asymmetric, or non-convex geometries. Authors in[30] employ Gaussian Process Regression to model traversability and plan smooth paths via gradient optimization over a continuous traverability map. Their model, however, captures only geometric aspects (terrain, obstacles), ignoring semantic labels.

Despite the advances, most of the approaches rely on discretization or heuristic weighting schemes, which oversimplify spatial distributions of objects. In contrast, our method projects semantic labels using Probability Density Functions (PDFs). This probabilistic treatment preserves uncertainty while retaining rich geometric and semantic detail. As a result, our representation enables more informative and robust cost maps for planning, addressing the limitations of rigid label maps.

Genetic algorithms in robotics

Genetic algorithms have consistently demonstrated their effectiveness as robust optimization techniques in addressing various robotics problems, particularly in path planning. Their capacity to adapt to different spatial environments and explore large solution spaces makes them especially suitable for dynamic and complex scenarios. Several works have proposed genetic algorithm methods for collision-free path planning in 2D maps considering complex scenarios[31,32]. Authors in[33] applied the Differential Evolution algorithm for path planning in grasping. This problem is particularly complex as it must integrate the geometric constraints of the kinematic chain of both the robotic hand and the arm. This highlights the versatility of evolutionary algorithms in addressing multi-constraint problems in robotics.

Despite their demonstrated effectiveness in global optimization tasks, genetic algorithms have yet to fully explore the integration of semantic data. To our knowledge, no prior work has applied semantic data to genetic algorithms for robotics tasks. Current implementations focus primarily on geometric or kinematic constraints, overlooking the rich contextual data provided by semantic features. Our method addresses this gap, enhancing robot navigation by solving issues like geometric redundancies through a more context-aware approach. By incorporating semantic information into the optimization process, we mitigate challenges such as geometric redundancies and improve the robot's context-awareness.

Semantic mapping with static objects

We propose the construction of a metric-semantic map $\mathscr{M}$ composed by two main elements: a global 2D occupancy grid map and a set of 3D rooms. Mathematically, this can be defined as:

$$\mathscr{M} = \{M_{\text{global}}, \mathscr{H}\} \tag{1}$$

where M_{global} is the 2D occupancy grid map and $\mathscr{H}$ is the set of 3D rooms.

The occupancy grid map is defined as:

$$M_{\text{global}} = \{(x, y, o) \mid x, y \in \mathbb{R},\ o \in \{0, 1\}\} \tag{2}$$

where (x, y) are spatial coordinates in the 2D plane and o indicates the probabilistic value of a cell being occupied ($o = 1$) or free ($o = 0$).

The set of rooms is defined as:

$$\mathscr{H} = \{H_i \mid i = 1, 2, \ldots, N\} \tag{3}$$

where each room H_i is defined by the following elements:

- A coordinate frame (x_i, y_i, θ_i) expressed with respect to the coordinate axis of the global 2D map, where (x_i, y_i) is the position of the origin of the room and θ_i is the orientation of the room relative to the global map.
- A polygon P_i defining the perimeter of the room's walls, represented by a set of vertices in the 2D plane:

$$P_i = \{(x_{i1}, y_{i1}), (x_{i2}, y_{i2}), \ldots, (x_{im}, y_{im})\} \tag{4}$$

- A set of labeled point clouds $\mathscr{O}_i$, each corresponding to a segmented static object in the room:

$$\mathscr{O}_i = \{O_{ij} \mid j = 1, 2, \ldots, K_i\} \tag{5}$$

where each labeled point cloud O_{ij} is defined as:

$$O_{ij} = \{(x, y, z, l) \mid x, y, z \in \mathbb{R},\ l \in \mathbb{L}\} \tag{6}$$

where (x, y, z) are the 3D coordinates of a point on the object and is the semantic label identifying the type of object, belonging to a set of labels $\mathbb{L}$.

2D occupancy grid map construction

The construction of the 2D occupancy grid map is achieved by leveraging data from a 2D LiDAR sensor mounted on the robotic platform. This sensor continuously measures the distance to surrounding obstacles, providing high-resolution range data of the immediate environment. The raw LiDAR measurements are then processed by a Simultaneous Localization and Mapping (SLAM) algorithm, which incrementally integrates the data to build a spatial representation while simultaneously estimating the robot's pose within the environment.

Through continuous updates and fusion of LiDAR scans, the occupancy grid evolves to accurately reflect the spatial layout of the surroundings. The resulting grid is represented as a discrete 2D grid where each cell encodes the probability of occupancy, distinguishing free space from obstacles.

3D room reconstruction

A point cloud reconstruction process is performed in each of the rooms. Open3D's reconstruction system[34], based on the work presented in[35], is applied. Starting from a recorded RGBD image sequence divided into subsets, camera poses are estimated between consecutive frames and submaps are extracted. Then, adjacent submaps are registered using Iterative Closest Point (ICP)[36] and distant ones are compared using geometric features to detect loop closures, constructing a pose graph in the global coordinate system which is optimized to produce the final 3D reconstruction.

Alternative reconstruction techniques exist, such as the approach presented in[37], which employs a 3D LiDAR sensor to map the entire environment and reconstruct it using an evolutionary algorithm. However, the method chosen in this work relies on RGB-D camera data due to its greater affordability and reduced execution time.

Semantic segmentation of 3D objects

To understand the objects present in each room, a semantic segmentation process is necessary. The procedure is performed using Minkowski Engine[38] trained with the ScanNet dataset[39], which focuses on indoor environments, specifically static objects that remain unchanged. Fig. 1 shows a representative example of the results for the reconstruction and segmentation of a room.

We assume that the reconstruction and semantic segmentation processes are sufficiently robust, as they rely on widely used and well-established methods. These processes could also be supported by ground truth data, since the involved features are considered stable over time. The reconstruction and labeling steps fall outside the scope of this paper, so our focus is primarily on leveraging the foundation they provide.



Fig. 1. Room reconstruction using an RGBD sequence and its semantic segmentation. Visual-SLAM is applied to reconstruct the environment in the first place. Then, the resulting point cloud is segmented. The result is a subset of labeled point clouds corresponding to commonly found static household objects. The figure was generated by the authors using raw images from a RealSense camera on a real scenario and Python's Open3D library[34].

Pose alignment and room registration

The 2D global map provides a planar representation of the environment, while the 3D point clouds capture detailed spatial and semantic information. Proper alignment between these two data sources is necessary to integrate their information effectively and improve the overall spatial understanding.

The semantically segmented information is also used to refine the local clouds alignment with respect to the 2D global map. Initially, the pose from which the data capturing process started serves as an estimate of the submap pose. Then, the floor plane (identified by points labeled as 'floor') is aligned with the horizontal global plane using a plane fitting operation to correct the inclination of the cloud. Next, points labeled as 'wall' are projected onto the floor plane and an ICP algorithm is applied to adjust any displacement on the horizontal plane. Fig. 2 depicts the resulting information extracted from a reconstructed room, where the relationship between the room information and the global map is highlighted.

Once the local cloud is correctly aligned, the segmented object point clouds are now accurately positioned within the corrected pose. This alignment allows for establishing the relationship between the coordinate axis of the local cloud and the 2D occupancy grid map. With the correct pose alignment, the perimeter of the room can be extracted. This is achieved by using the points labeled as walls, which have been projected onto the floor plane. These projected points are then used to delineate the room's boundaries. By identifying the outermost points of the projected wall data, the perimeter of the room is defined. Thus, with the objects' point clouds in their correct poses, the room perimeter clearly defined and the corresponding relationship between the local and the global reference frame, the metric-semantic map $\mathscr{M}$ is enriched with semantic information.

Probabilistic treatment of labeled objects

Each labeled object instance observed in the 3D environment is represented as a continuous spatial distribution over the 2D floor plane. Instead of assigning rigid geometric primitives, the spatial footprint of each object is encoded using a Probability Density Function (PDF) constructed from the projected 3D points associated with that instance.

The procedure operates by first isolating the points corresponding to a specific object category and filtering them by spatial containment within each room polygon. This masking ensures that probabilistic information does not propagate across walls, which typically delimit semantically distinct regions. Within each polygonal region, the method applies unsupervised clustering to the object points using the DBSCAN algorithm[40], which identifies clusters based on point density without requiring prior knowledge of the number of instances. Each resulting cluster is assumed to correspond to a distinct object instance.

For each identified cluster, a PDF is estimated via kernel density estimation (KDE). The KDE is computed over a 2D spatial grid aligned with the environment map, where each cell corresponds to a physical location. The bandwidth of the kernel is adaptively scaled to control the smoothness of the resulting distribution. The estimated density is normalized per object to ensure consistent contribution across instances, and spatially masked to restrict support to the corresponding room polygon.

Each probability layer for a semantic category is obtained by summing the masked density contributions of all individual object instances within the environment. The final probabilistic map then maintains separate layers for each object category, ensuring that information from different classes is preserved without overwriting and instead contributes in a complementary manner. This approach yields a smooth and spatially aware representation of object likelihoods, which is particularly useful for downstream tasks such as path planning or localization under semantic priors.

Divergence-based semantic pose optimization

This section defines the first proposed application of the probabilistic semantic maps: a relocalization system based on the comparison between the global map and the local information captured by the robot. For this task, the DE optimization method has been selected. First, the procedure on how the robot captures local semantic information is shown. Then, the adaptation of the DE algorithm to the proposed problem is explained. Finally, the selected cost function is described in more detail.

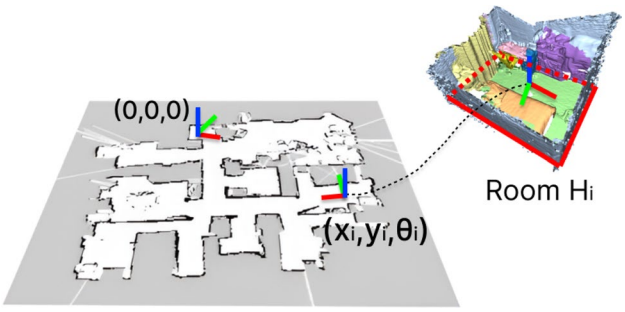


Fig. 2. Correspondence between a room H_i and the global occupancy grid map. The global map shows its reference frame at (0,0,0), whereas room H_i is represented with a set of labeled point clouds, its origin at (x_i, y_i, θ_i) and its perimeter P_i marked in red. The figure was generated by the authors using Python's Open3D library[34].

Partial view capturing

The DE cost function is based on comparing the global map and the objects locally detected by the robot. This requires the robot to have an RGBD camera horizontally mounted on it. In our case, the camera model used is the RealSense D-435i. This camera model publishes a point cloud aligned with the image through its ROS package, where each point has a direct correspondence with the image pixels.

From this information, we propose the extraction of a partial view representation of the environment. We define a partial room map H_{loc}, which contains a set of segmented data where each label represents a detected object type. Mathematically, H_{loc} is defined as:

$$H_{\text{loc}} = \{O_n \mid n = 1, 2, \ldots, K\} \tag{7}$$

where each O_n represents the point cloud of a segmented object type, defined by:

$$O_n = \{(x, y, z, s) \mid x, y, z \in \mathbb{R},\ s \in \mathbb{S}\} \tag{8}$$

In this context, (x, y, z) are the 3D coordinates of points in the cloud, and s is the semantic label identifying the object type, belonging to a set of labels $\mathbb{S}$. Although the notation resembles that used in Eq. 6, it should be noted that the label set $\mathbb{L}$ defined there corresponds to a different dataset, and thus $\mathbb{S}$ and $\mathbb{L}$ are not interchangeable.

When visual data is obtained from the robot pose, YOLOv8[41] trained on the COCO dataset[42] is applied on the image to obtain object masks. Then, 3D points corresponding to each type are extracted from the point cloud. The result is a set of point clouds in the robot reference frame, where each cloud corresponds to each of the detected object types. That is, if for example three chairs were detected, only one point cloud would be extracted, since they belong to the same object type.

Differential evolution

The Differential Evolution (DE) algorithm[43] is an optimization method based on genetic principles to iteratively refine a population of candidate solutions. DE proceeds iteratively by evolving a population of N candidate solutions $\{\mathrm{x}_i^g\}_{i=1}^N$ over generations g through three main operations: mutation, crossover, and selection. Regarding mutation, for each individual x_i^g, a mutant vector v_i^g is generated by combining three other distinct individuals:

$$\mathrm{v}_i^g = \mathrm{x}_{r1}^g + F \cdot (\mathrm{x}_{r2}^g - \mathrm{x}_{r3}^g) \tag{9}$$

where $r1$, $r2$, $r3$ are distinct indices and $F \in [0, 2]$ is the mutation factor.

For crossover, a trial vector u_i^g is created by mixing the target vector and the mutant vector:

$$u_{i,j}^g = \begin{cases} v_{i,j}^g & \text{if } \mathrm{rand}_j < C_r \text{ or } j = j_{\text{rand}} \\ x_{i,j}^g & \text{otherwise} \end{cases} \tag{10}$$

where $C_r \in [0, 1]$ is the crossover rate, and j_{rand} ensures at least one component is inherited from the mutant.

In the selection phase, the trial vector replaces the original if it results in a lower cost:

$$\mathrm{x}_i^{g+1} = \begin{cases} \mathrm{u}_i^g & \text{if } C(\mathrm{u}_i^g) < C(\mathrm{x}_i^g) \\ \mathrm{x}_i^g & \text{otherwise} \end{cases} \tag{11}$$

where $C(\mathrm{x})$ is the cost function. This evolutionary process continues for a predefined number of generations or until convergence.

The adaptation of the DE algorithm for the relocalization proposal of this work is presented in Fig. 3. In the context of visual relocalization, we aim to find the best candidate robot pose $\mathrm{x} = [x, y, \theta]^T$ that aligns the current observation with the precomputed semantic map. This can be formulated as minimizing a cost function $C(\mathrm{x})$ that quantifies the dissimilarity between the current scene and the hypothesized pose, such as Kullback-Leibler divergence between probability distributions.

The N individuals representing robot poses are evenly distributed over free areas to initially evaluate all the possible locations where the robot could be placed. Individuals are mutated and recombined to generate a set of new candidate poses, which are evaluated in the selection process. For this evaluation, the local room map H_{loc} that the robot is observing is translated and rotated to match the candidate robot pose and it is compared against the global map $\mathscr{M}$. If the cost of the candidate robot pose is smaller than the cost of its parent, the candidate pose is selected for the new generation. Otherwise, the parent pose is maintained. This evaluation is detailed below. The explained process is repeated for the total number of specified iterations.

Probability-based cost function

To evaluate the alignment between a partial observation and the global map, we compute a cost function based on the divergence between their respective spatial distributions. For each detected object class, a cost is computed and the final score is the average across all classes. The process is illustrated in Fig. 4, and detailed below.

Initially, H_{loc} is transformed to align with the hypothesis pose. Before computing any cost, two geometric constraints are applied to discard implausible hypotheses. We assume that observed objects must lie in the same room as the robot, without walls between them. To enforce this, we compare the room perimeter of the candidate pose with that of the detected objects using the Intersection over Union (IoU) metric[44]; poses with IoU below 0.9 are discarded. Additionally, if the line segment connecting the robot to the object centroid crosses any

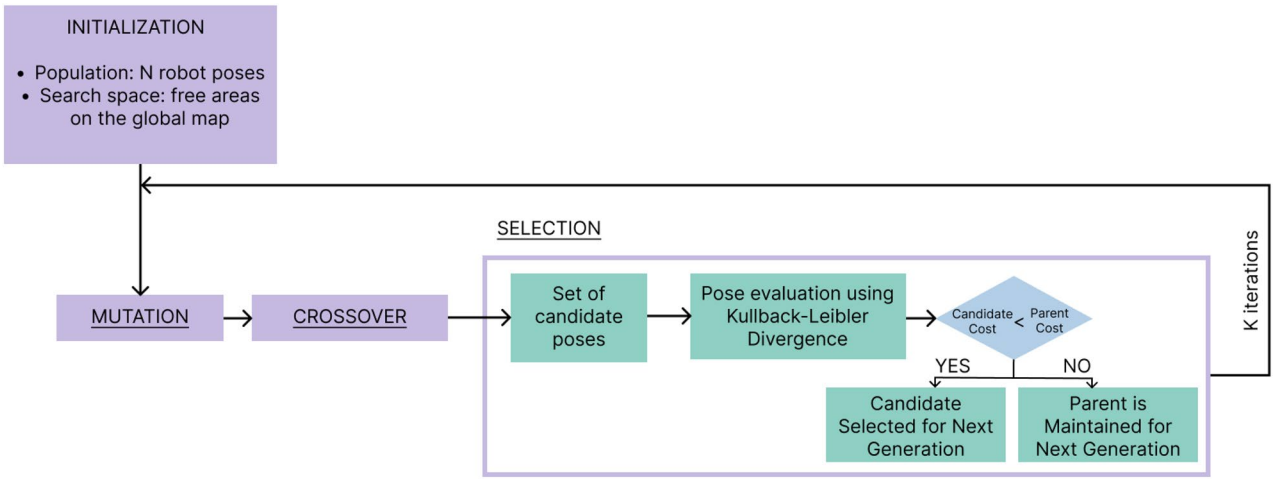


Fig. 3. DE adaptation to the relocalization problem. The population represents robot poses in the global map. In the selection process, the comparison between the partial room map H_{loc} and the metric-semantic map $\mathscr{M}$ is performed.

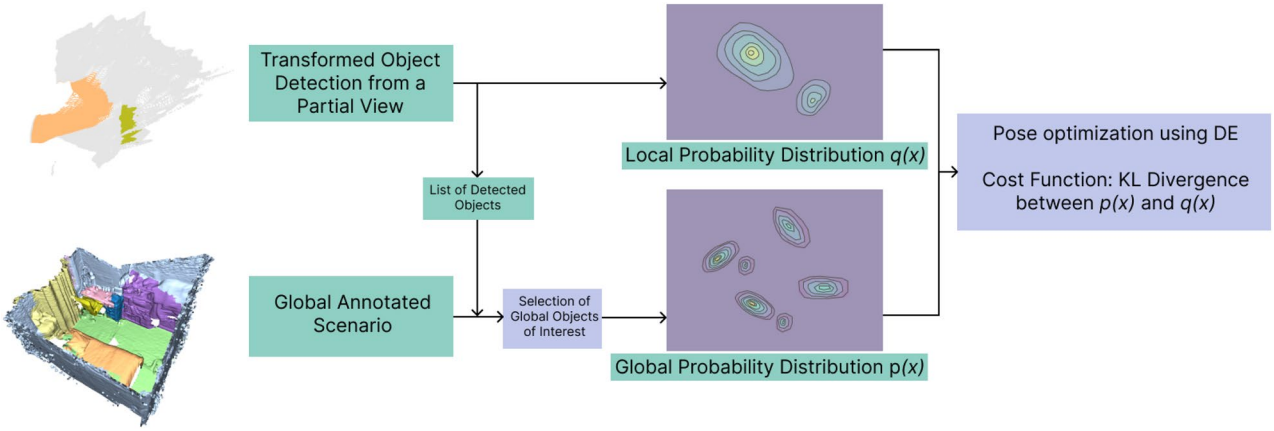


Fig. 4. Probability-based comparison between the transformed partial view and the global metric-semantic map. Objects are projected to the ground plane and characterized by a probability distribution. The Kullback-Leibler Divergence function estimates the similarity between the two distributions to assign a cost to the evaluated pose.

room boundary, the hypothesis is rejected. Finally, if the room corresponding to the candidate pose lacks any of the detected object classes, the pose is also discarded. Note that this room-based filtering is not meant to localize within each room independently. Instead, room segmentation is used as a spatial constraint to reduce the search space, improving efficiency and focusing computation on semantically consistent regions.

If these conditions are satisfied, the pose is evaluated. The local point cloud is projected onto the ground plane, downsampled, and modeled using a probability density function (PDF) via SciPy's *gaussian_kde*. This function estimates a continuous spatial distribution of object presence by averaging Gaussian kernels centered at each observed 2D point $\{(x_k, y_k)\}_{k=1}^{n}$:

$$\hat{f}(x,y) = \frac{1}{n}\sum_{k=1}^{n} K_h(x - x_k, y - y_k) \tag{12}$$

where K_h is a smoothing kernel, typically Gaussian:

$$K_h(x,y) = \frac{1}{2\pi|h|^{1/2}} \exp\left(-\frac{1}{2}\begin{bmatrix}x\\y\end{bmatrix}^T h^{-1}\begin{bmatrix}x\\y\end{bmatrix}\right) \tag{13}$$

where h is the covariance matrix or bandwidth of the kernel. This matrix can be automatically calculated from the provided data using Scott's Rule[45].

A matching process is then applied: for each object class in the local view, the corresponding points in the global map are processed in the same way to estimate a reference PDF. The similarity between the local distribution $\hat{q}$ and the global one $\hat{p}$ is quantified using the Kullback-Leibler divergence:

$$D_{KL}(\hat{q}||\hat{p}) = \sum_i \hat{q}(x_i) \log\left(\frac{\hat{q}(x_i)}{\hat{p}(x_i)}\right) \tag{14}$$

The order of the distributions is crucial in assessing the Kullback-Leibler divergence, as it is not symmetric. This measurement evaluates the difference between two probability distributions, $\hat{q}$ and $\hat{p}$, indicating how much the partial view distribution deviates from the full map distribution.

Semantically biased path planning

To generate navigation behaviors that align with human-centric preferences, we construct a flexible cost map that integrates semantic object distributions, obstacle occupancy, and safety margins. This semantically biased path planning framework enables the combination of multiple probabilistic layers, each representing a spatial preference derived from object-level information.

Each semantic layer encodes the likelihood of presence for a specific object category (e.g., beds or tables) using per-instance Gaussian representations as previously described. These layers can be assigned positive or negative weights depending on their desirability: for instance, tables may be treated as attractors (positive weights), while beds may be considered repellers (negative weights). These preferences are encoded by scaling each object distribution with a sign-adjusted factor prior to fusion. Importantly, multiple semantic layers can be superimposed without restriction, allowing the planner to simultaneously favor and avoid different object categories.

In addition to semantics, we incorporate an occupancy-derived cost map to ensure obstacle avoidance. This map is computed from a dense point cloud representing the environment layout, where free and occupied regions are inferred from color or geometry cues. As a final safety enhancement, a distance transform is applied to the occupancy map, and a repulsive cost is added in regions close to walls and other obstacles. This penalizes trajectories that come too close to structural boundaries, promoting safer paths that maintain clearance.

All cost components are fused into a unified cost map. Semantic adjustments are only applied in regions marked as free space, and the resulting map is normalized to ensure compatibility with grid-based path planners. We then employ a graph-based planning algorithm (A*) on the inverted cost surface to compute trajectories from start to goal locations. This results in paths that are not only feasible and safe, but also aligned with task-relevant spatial preferences.

While we demonstrate the approach using A*, the resulting cost map is compatible with a wide range of planning algorithms that operate over scalar cost fields. In particular, sampling-free methods such as Fast Marching Square (FM2)[46] can directly leverage the continuous nature of the fused map, allowing for smoother and more globally consistent trajectories.

Experimental results

To evaluate the effectiveness and versatility of the proposed semantic representation, we test it in two fundamental robotic tasks: global relocalization and path planning. The goal is not only to assess performance in these applications, but also to demonstrate that a single, unified map structure can support multiple functionalities without requiring task-specific representations. This highlights the potential of our approach to serve as a general-purpose semantic layer for robotic decision-making.

Experiments were conducted in two distinct scenario types: a benchmark dataset (the Replica dataset[47]) and a real environment. The real environment consists of an apartment of approximately 90 square meters, including three bedrooms, two living rooms, two bathrooms, and a kitchen. The inherent complexity of the real environment allows for the evaluation of the effectiveness of our localization method under realistic conditions. The robot used in the experiments was a TurtleBot2, equipped with a 2D Hokuyo laser sensor and an Intel RealSense camera.

Localization results

The main objective of this experiment is to evaluate the method's localization capabilities. A series of tests have been carried out using both a commonly used dataset and a real environment, in order to compare the results obtained with those from state of the art localization techniques. It is important to note that our approach performs global optimization without assuming any prior knowledge of the robot pose, unlike many methods in the literature which rely on an initial pose estimate (e.g., from GPS or odometry).

Before conducting the experiments, a fine-tuning process was performed on the parameters of the DE algorithm to optimize both the number of particles and iterations required. This parameter adjustment is aimed to enhance the performance and convergence speed of the proposed approach by generalizing the parameter values.

Differential evolution fine-tuning

The first step in the experimental evaluation involves fine-tuning the parameters of the Differential Evolution algorithm to ensure optimal performance in subsequent experiments. This process is essential to balance computational efficiency and localization accuracy, as the choice of parameters directly impacts the algorithm's convergence and effectiveness. Specifically, we focused on adjusting the number of particles and the number of iterations, as these parameters play a critical role in determining the algorithm's ability to explore and exploit the

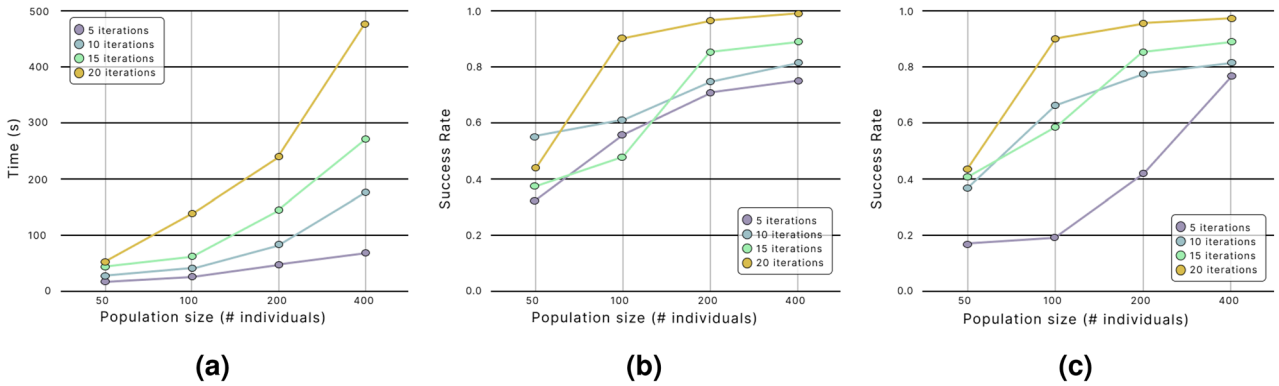


Fig. 5. Differential Evolution evaluation with respect to population size and number of iterations using three metrics: (a) execution time, (b) the success rate of correctly localizing the robot in the correct room and (c) the success rate of accurately determining the robot's pose.

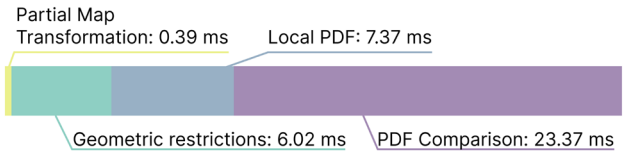


Fig. 6. Time analysis for the evaluation of the cost function, where the implied processes are shown in order of execution. The mean total time is 37.15 ms for one execution.

solution space effectively. The results of this fine-tuning provide the foundation for selecting the most suitable configuration for the experiments described later.

The performance of DE was evaluated by varying two key parameters: the population size (50, 100, 200 and 400 particles) and the number of iterations (5, 10, 15 and 20). The evaluation was conducted using three metrics: execution time, the success rate of correctly localizing the robot in the correct room and the success rate of accurately determining the robot's pose within the correct room. Results are shown in Fig. 5.

The first graph shows that execution time increases proportionally with both population and iterations. In the second graph, it can be seen that increasing the population size significantly improves the success of localizing the robot in the correct room, but increasing the number of iterations is not as effective. The third graph highlights similar trends for pose localization, with a higher impact for the number of iterations. This data suggests that it is relevant to set a sufficiently large population size while having enough iterations for exploiting this data to find an optimal solution.

It must be considered that while larger populations and more iterations improve accuracy, they significantly increase computation time, which may be impractical. Based on these observations, the combination of 200 particles and 20 iterations provides a balance between computational efficiency and localization accuracy, as it achieves high success rates while avoiding the steep time costs associated with larger populations. In the following sections, these parameters are kept constant.

Runtime analysis

Regarding the computation time required to execute the proposed relocalization method, it is important to note that this duration is influenced by the chosen DE parameters and the number of observed objects. Approximately, two minutes and a half are required by the algorithm to locate the robot on the built map for 20 iterations with 200 particles, measured on a 12th Gen Intel Core i7-12700H CPU. A more detailed time evaluation is shown in Fig. 6, where the required time for each relevant process on the cost function is measured. The calculation of the global PDF is not shown, as it is only calculated once before the DE algorithm starts.

As it can be seen, the most time-consuming operations are the local PDF computation and, especially, the PDF comparison. The comparison of two PDFs using Kullback-Leibler divergence is particularly slow due to the need to evaluate the probability distributions at every point on the map. As the objects shapes become more detailed when using PDFs compared to other methods, the computational cost increases. However, this additional time investment leads to improved success rates, as will be discussed in the following section.

As a comparative evaluation against other global localization approaches, we provide a comparison with the following methods: Adaptive Monte Carlo Localization (AMCL), Particle Swarm Global Localization (PSGL), Genetic Algorithm Global Localization (GAGL), Differential Evolution Global Localization (DEGL), and Perfect Match Global Localization (PMGL). These data were obtained with similar parameter settings and are reported in a comparative study[48]. The results are summarized in Table 1.

	AMCL	PSGL	GAGL	DEGL	PMGL	Ours (DE + Semantic Map)
Avg. Time per Iteration [s]	0.02	0.07	0.03	0.04	1.63	0.04

Table 1. Average time per iteration (in seconds) for different global localization algorithms.

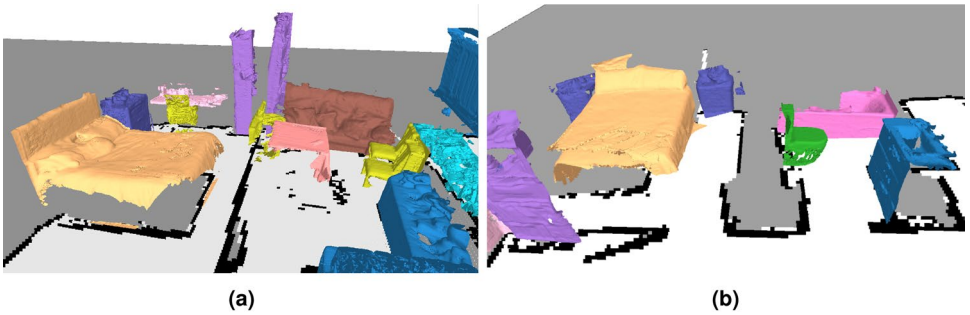


Fig. 7. Close-up view of the constructed metric-semantic map showing object point clouds aligned with the global occupancy grid map: (**a**) bedroom and living room (**b**) bedroom and toilet. The figure was generated by the authors using Python's Open3D library[34].

The results show that our method achieves competitive execution times compared to other global optimization algorithms while leveraging richer semantic information for more robust and accurate localization. In particular, compared to DEGL, our method maintains similar efficiency despite incorporating both semantic and geometric information, demonstrating that the inclusion of semantic knowledge does not significantly increase computational cost while enhancing localization robustness.

Results on the real scenario

In this section, we evaluate our method in a real-world environment, as opposed to the controlled conditions of a dataset. Real-world scenarios often introduce additional uncertainty and challenges such as sensor noise, dynamic objects and inconsistencies in environmental features, which can significantly impact the performance of localization methods. This evaluation aims to assess the robustness and adaptability of our approach under these more complex and unpredictable conditions, providing a comprehensive analysis of its applicability in practical scenarios.

We demonstrate the construction and utilization of our proposed metric-semantic map in a real-world indoor environment for relocalization. First, the map construction process is initialized. Using the TurtleBot2 equipped with a 2D Hokuyo laser sensor and a RealSense camera, we collect data to build the occupancy grid map of the entire environment. This global 2D occupancy map provides the foundation for our mapping process. Each of the eight rooms are then identified and mapped individually. For each room, the perimeter is delineated, and static objects within the room are segmented and represented as point clouds. The pose of each room, defined relative to the global coordinate system, is also extracted. Fig. 7 illustrates the constructed map, showing the global occupancy map augmented with labeled objects. As it can be seen, this map allows for a more intuitive and context-aware representation of the environment, where semantic labels play a major role.

Once the map is constructed, we test the relocalization capabilities of the proposed method on the robot. The robot is moved to a random pose to capture the surrounding information with the RGBD sensor. Fig. 8 depicts scenes captured by the robot, highlighting several challenges like partial occlusions, non-unique objects and dynamic elements. Next, the global robot pose is calculated using DE. The PDFs extracted from the partial view on each individual and the global map are compared using the Kullback-Leibler divergence to determine their similarity. The process is repeated for each object instance and the obtained distributions are averaged and normalized. This mean value is associated to the individual that is being evaluated to represent its cost.

While the averaging of object-wise KL divergences provides robustness, the method's performance inherently depends on the underlying object detection and classification quality. In scenarios where multiple objects are detected, the probabilistic nature of our approach allows it to tolerate individual misclassifications or false negatives, as the correct matches typically dominate the mean cost. However, a limitation arises when only a single, misclassified object is present, which could lead to an incorrect pose estimation. To mitigate this, our framework could be extended by incorporating a geometric consistency check or a recovery strategy, such as having the robot perform an in-place rotation, to gather additional semantic evidence and refine the global PDF, thereby increasing the confidence of the relocalization before finalizing the pose estimate.

We compare our approach in the real scenario against two widely applied localization methods: Adaptive Monte Carlo Localization (AMCL)[49] and ORB-SLAM2[50], which are available as ROS packages. This comparison allows us to benchmark our method against established standards, providing a comprehensive assessment of its performance in a real-world scenario. AMCL is a probabilistic localization technique that uses a particle filter to estimate the robot pose, while ORB-SLAM2 is a SLAM library used for computing camera trajectories.

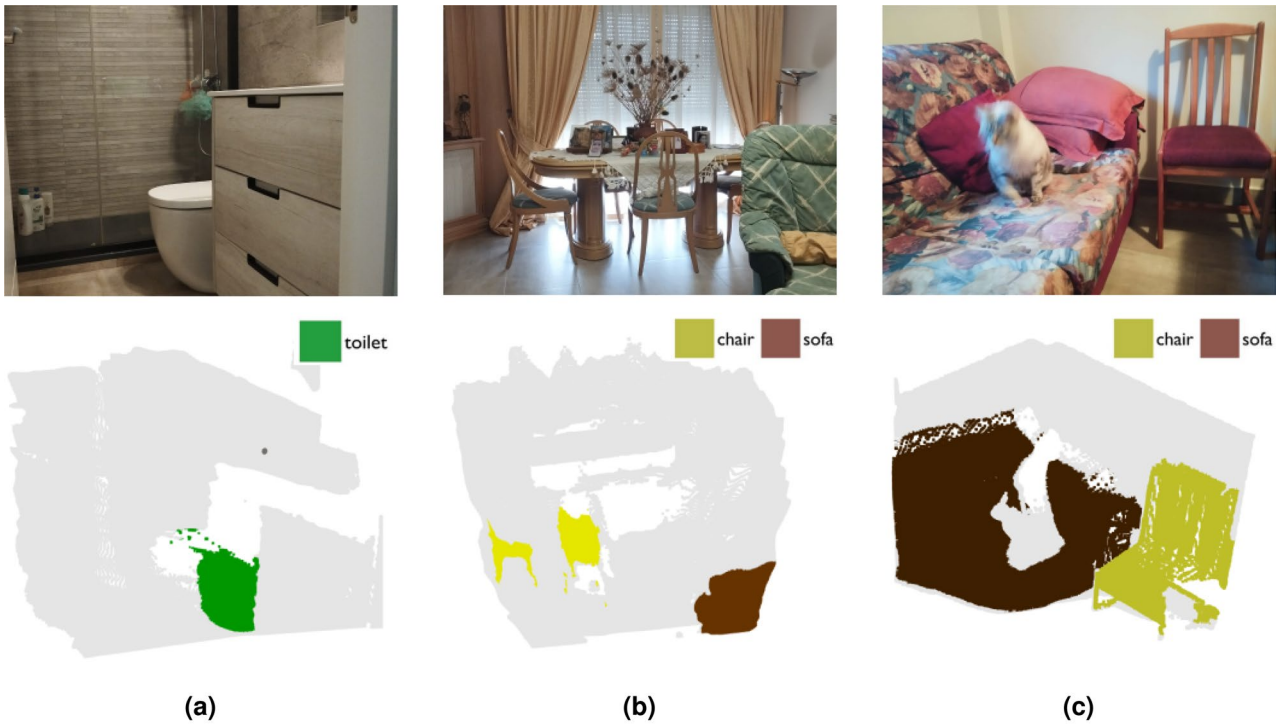


Fig. 8. Challenging object detections made at random poses, showing common real-world challenges: (**a**) occlusions in views with single objects, in this case an occluded toilet, (**b**) generic objects that are commonly found in several areas, like chairs and sofas, (**c**) dynamic elements like people or pets walking by, which may cause a variation in keypoints detection or occlusions. The figure was generated by the authors using raw images from a RealSense camera on a real scenario and Python's Open3D library[34].

Method	Success Rate	Error [m]	Error [rad]
AMCL[49]	0.58	0.177 ± 0.061	0.165 ± 0.048
ORB-SLAM2[50]	0.67	0.146 ± 0.071	0.065 ± 0.033
Ours	0.85	0.159 ± 0.052	0.074 ± 0.017

Table 2. Localization Results on the real scenario.

Experiments were performed applying the two above relocalization algorithms as well as our proposal. A total of 100 tests were performed, corresponding to 20 indoor locations executed five times each. The selected test poses were placed in characteristic areas of the environment, with variability of distinctive elements to facilitate localization. The aim is to compare the resulting robot poses with respect to the ground truth values, which are set by hand. The selected metrics are success rate (a pose is considered successful if it is under 0.3m and $\pi/8$ away from the ground truth), position error in meters and orientation error in radians. The two latter ones are calculated only in successful poses. Table 2 collects the results obtained by applying the three proposed algorithms.

As it can be seen, our method highly surpasses the other two methods regarding the success rate. AMCL could not handle repetitive regions like rooms given its limitations derived from only using geometric information, and ORB-SLAM2 repeatedly lost the camera odometry. Regarding errors in position and orientation, ORB-SLAM2 obtained the best mean error, closely followed by our method. Although DE is a flexible algorithm that can find solutions very close to the optimum, its stochastic nature and parameter dependence means that it is not always guaranteed to find the exact global minimum. However, the solutions are close to the optimum in complex search spaces such as the one proposed. Moreover, our deviation values were smaller, meaning that we have less variable and more consistent results.

Results on a benchmark dataset

The Replica dataset consists of 18 highly realistic 3D indoor scene reconstructions, providing a rich set of object instances across 88 different semantic categories. For this work, we focus on the apartment environments, as they offer the most semantic richness and introduce an additional level of complexity for global localization.

We compare the performance of our method against three state-of-the-art approaches from the literature, representing some of the most widely used techniques in the field: SLAM-net[18], C*[19] and NICE-SLAM[25]. Few

Method	Success Rate	Error [cm]
SLAM-net[18]	78.80%	17.00
C*[19]	49.70%	19.50
NICE-SLAM[25]	89.33%	**2.85**
Ours	**91.07%**	10.29

Table 3. Localization Results on the Replica dataset.

	Semantic Cost			Length			Smoothness		
	A*	Gauss-A*	PDF-A*	A*	Gauss-A*	PDF-A*	A*	Gauss-A*	PDF-A*
Positive	0.6064	0.5227	0.5197	1.5054	1.6113	1.6018	0.1163	0.1578	0.1753
Negative	0.3546	0.2373	0.2189	1.5482	1.6378	1.6615	0.1032	0.1961	0.1315
Mixed	0.4205	0.3783	0.3377	1.4785	1.5372	1.5779	0.0975	0.1262	0.0890
Mean	0.4488	0.3607	**0.3374**	**1.5104**	1.5949	1.6134	**0.1054**	0.1575	0.1271

Table 4. Path planning results on the real scenario.

	Semantic Cost			Length			Smoothness		
	A*	Gauss-A*	PDF-A*	A*	Gauss-A*	PDF-A*	A*	Gauss-A*	PDF-A*
Positive	0.4909	0.4839	0.4445	1.0548	1.0654	1.1957	0.0878	0.1653	0.0999
Negative	0.3524	0.2637	0.2653	1.1746	1.2344	1.177	0.1657	0.3577	0.2355
Mixed	0.2378	0.2145	0.1896	1.0812	1.0718	1.1686	0.1025	0.1137	0.1009
Mean	0.3452	0.3014	**0.2817**	**1.1024**	1.1212	1.1804	**0.1142**	0.1596	0.1578

Table 5. Path planning results on the benchmark scenario.

existing methods tackle the problem of fully global, pose-agnostic localization, as most rely on an initial pose estimate and local optimization. Therefore, our approach targets a scenario that is less frequently addressed in the literature, and the selected comparison set reflects methods that are suitable references for this global relocalization task. The selected comparison metrics are success rate (a pose is considered successful if it is under 0.3m and $\pi/8$ away from the ground truth) and root mean square (RMS) position error. The latter one is calculated only in successful poses. Results are shown in Table 3.

Results demonstrate that our method significantly outperforms feature-based approaches like SLAM-net and C* in both success rate and RMSE. In comparison to the implicit neural field-based method, NICE-SLAM, our approach achieves a very similar success rate, although with a higher RMSE. This difference is expected, as evolutionary algorithms like DE, while robust at avoiding local minima, do not always find the absolute minimum because they are stochastic. Nonetheless, these results highlight the validity of our method, offering a reliable alternative where the map used for robot localization can be leveraged for additional applications, unlike neural field-based approaches, which are typically challenging to extrapolate beyond specific tasks. Furthermore, SLAM-net's reliance on deep learning for feature extraction and matching limits its generalization to novel environments, while C*'s graph-based optimization, though effective in refining poses, struggles with achieving a comparable success rate, particularly in complex scenarios.

Path planning results

To evaluate the impact of the semantic representation on path planning, we compare three configurations: classical A* over the binary occupancy map (A*), A* guided by a Gaussian-based semantic layer (Gauss-A*), and A* using the full per-object probabilistic density functions (PDF-A*). Each method is applied over the same environment and start-goal pairs.

For every resulting path, we compute three evaluation metrics: semantic cost, relative length, and smoothness. The semantic cost corresponds to the accumulated cost values along the trajectory over the semantic map, normalized by the path's total length to ensure comparability. Relative length measures the ratio between the actual path length and the Euclidean distance between start and goal, thus reflecting geometric efficiency. Lastly, smoothness is quantified as the mean absolute variation in direction between consecutive path segments, offering a compact measure of path continuity and navigability.

A total of 90 experiments were conducted in each scenario, dividing the tests into three groups: maps with only positive biases, maps with only negative biases, and maps with mixed biases. This separation aims to analyze how the sign and combination of semantic biases affect the overall path planning performance. Tables 4 and 5 summarize the quantitative evaluation of the three path planning approaches on the real and the benchmark scenario, respectively.

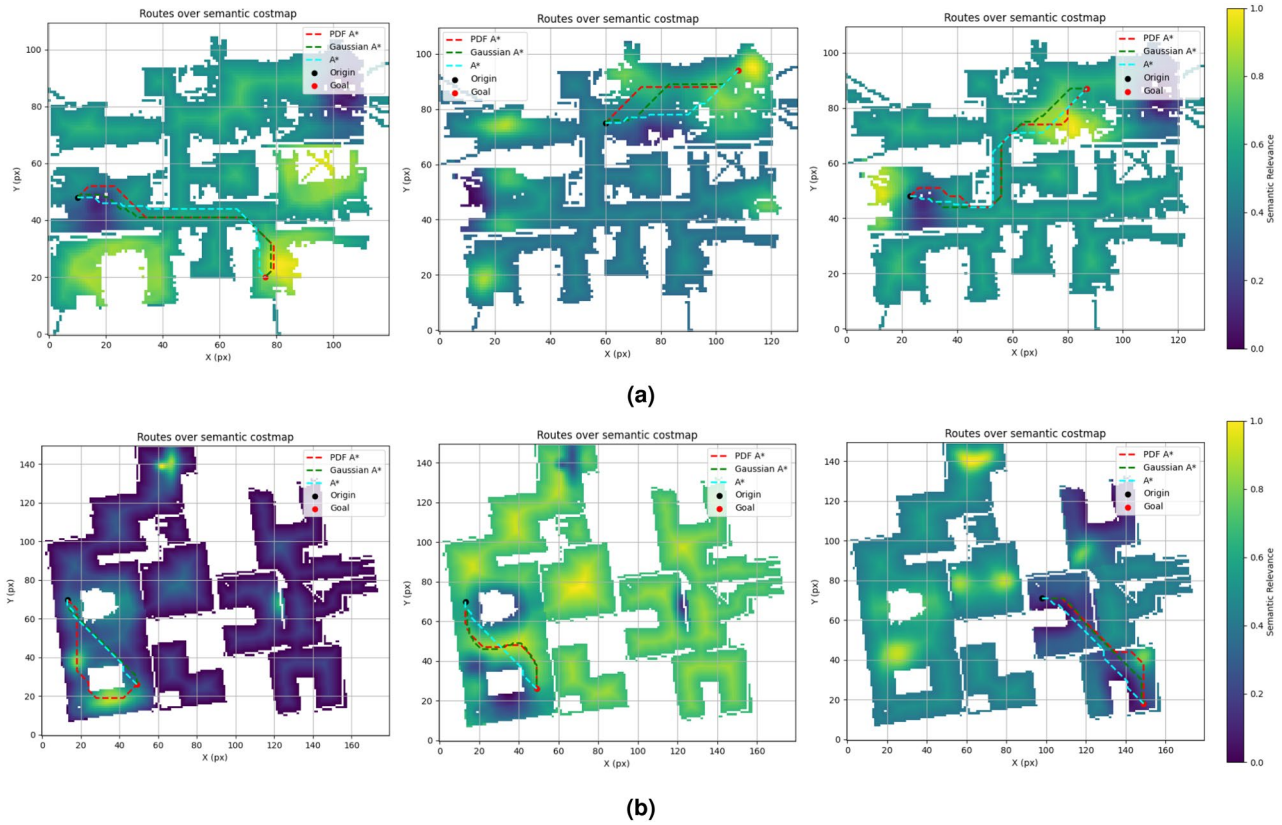


Fig. 9. Qualitative comparison of planned paths. (**a**) Paths in the real scenario. (**b**) Paths in the benchmark environment. In both cases, paths generated with PDF-A* better follow the semantic preferences encoded in the probabilistic map and exhibit higher smoothness compared to baseline methods.

In both scenarios, the PDF-A* method consistently achieves the lowest average semantic cost, indicating that paths planned using the full probabilistic density functions better adhere to the semantic constraints encoded in the map. Gaussian A* also reduces semantic cost compared to the baseline A*, though to a lesser extent, suggesting that the richer semantic representation in PDF-A* is more effective for biasing path planning.

Regarding path length, the classical A* algorithm tends to produce the shortest routes, as expected, since it only considers obstacle avoidance without semantic preferences. The semantic methods, particularly PDF-A*, generate slightly longer paths due to the influence of semantic biases, which may encourage detours to favor or avoid specific zones.

Smoothness results vary between scenarios. In the real scenario, PDF-A* yields the smoothest paths on average, potentially reflecting that semantic information helps avoid abrupt changes in direction by guiding the path through semantically meaningful corridors. Conversely, in the benchmark scenario, A* has the lowest smoothness metric (i.e., smoother paths), possibly because the semantic layers introduce more complex cost landscapes, leading to more direction changes.

Overall, these results demonstrate that integrating probabilistic semantic information into path planning via PDFs improves adherence to semantic preferences at the cost of increased path length and, depending on the environment, changes in path smoothness.

Figure 9 illustrates qualitative examples of planned paths in both the real and benchmark scenarios. The paths generated using the PDF-A* method demonstrate better alignment with the underlying semantic probabilities of the environment, favoring or avoiding regions according to their encoded semantic significance. Compared to the standard A* and Gauss-A*, the PDF-A* paths adhere more closely to the semantic cost structure with fewer abrupt turns. This supports the quantitative findings and highlights the advantages of using a full probabilistic representation for guiding path planning.

Conclusions

In this paper, we proposed a unified metric-semantic map for indoor environments that combines a 2D occupancy grid with probabilistic semantic information about object locations. Rather than designing a separate representation for each robotic function, our aim is to provide a single, expressive map that can support multiple tasks without additional adaptation.

We demonstrated the utility of this map in two core robotics applications: localization and path planning. For localization, we employed Differential Evolution to align partial observations with the map, using Kullback-Leibler divergence as a similarity metric. Experiments on both benchmark data and a real robot platform showed

that our approach achieves high accuracy, outperforming AMCL and ORB-SLAM2 particularly in ambiguous or occluded environments. For path planning, we evaluated the map's use with several planners, including a probabilistic variant that exploits the semantic distributions. The results showed improved alignment with semantic preferences and smoother paths, while maintaining competitive path lengths. Overall, these results validate the flexibility and effectiveness of using a common representation across different robotic capabilities.

For future work, we aim to integrate our global relocalization approach with efficient local tracking frameworks. Specifically, once an initial pose is estimated, a laser-based tracking method could be employed to maintain the robot's pose with low computational overhead. Our semantic localization method would then serve as a robust recovery mechanism, re-initializing the system if the tracking error exceeds a certain threshold or if the robot becomes lost. Additionally, we plan to further explore planning and interaction strategies that exploit the semantics of the map, including grounding user instructions via natural language.

Data availability

The dataset used during the current study is available from the corresponding author on reasonable request.

Received: 6 August 2025; Accepted: 13 February 2026
Published online: 17 February 2026

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Author contributions

A.Mo., L.M. and R.B. conceived the experiments, A.Mo. developed the software and conducted the experiments, and A.Mo. and A.Me. analysed the results. All authors reviewed the manuscript.

Funding

This research was supported by the project Advanced Mobile dual-arm manipulator for Elderly People Attendance (AMME) (PID2022-139227OB-I00), funded by Spanish Ministerio de Ciencia e Innovacion.

Declarations

Competing Interests

The authors declare no competing interests.

Additional information

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